Smart Helmet Using RF and WSN Technology for Underground Mines Safety

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***Abstract*— The underground mines all over the world are adversely affected by various hazards including gaseous explosions, landslides, fire hazards, etc. This leads to the significance of safety for the mine workers. A better communication technology has to be employed for an intelligent sensing and warning system. For this, RF technology is chosen for the communication inside the mines. The wireless sensor network is provided for the sensing of adverse working environment conditions. Combining both these technologies, a new smart helmet module is developed. The wireless sensor network consisting of various sensors senses mine environment parameters like temperature, pressure, humidity, gases like methane, carbon monoxide, etc. The measured parameters are processed and are used for an early intelligence warning system with the help of programmed alarm sounds, if there is any serious deviation from the normal parameter value range. The RF technology is also used as locating system, which helps in locating the mine-workers. This RF based wireless sensor network is reliable with easy installation and fast sensing and locating system. The design provides three modules, namely helmet module, localizer module, and control room module. The helmet module comprises of various sensors, microcontroller, antennas, encoders, decoders, etc. The helmet module acts as an intelligent, low power node in the wireless sensor network. The control room module contains antennas, PC, decoder, etc. The sensors are used to measure the variations in the temperature, humidity, pressure, fire and take decisions for required actions based on the measured data to ensure the safety of the underground mine workers. The RF technology also ensures the localization of the mine workers for their safety in a dangerous working platform. Thus the proposed system ensures safety and a reliable wireless communication inside the underground mines.**

***Keywords*— RF, wireless sensor network, smart helmet, mobile node, localization**

I. INTRODUCTION

The underground mining environments across the world have distinct characteristics including topology and other physical and chemical features. As the main focus of the study is on the communication scheme inside the underground mines, the general conditions to be considered are underground topology, geological construction, line-of-sight, dielectric medium, gaseous environment, temperature and humid conditions. Amidst of these dynamically changing natural conditions, there arises a necessity to engineer a better communication scheme which can be an appropriate solution to the series of mine disasters all over the world. A careful design and implementation of the communication technology is required to save lives from the mine hazards.

The propagation of the electromagnetic waves through underground mines is influenced by path loss, multipath fading, reflection/refraction, reduced propagation velocity, noise, and realistic waveguide effect. The three modes of communication through the underground mines are Through-the-Wire, Through-the-Air, and Through-the-Earth.

The helmet is used to protect the user‘s head and here, the Smart helmet for the miners has additional features like sensing system, locating system, communication system, etc.

The Radio Frequency communication technology uses transmission of radio signals, having the frequency range 30 KHz-300 GHz, inside underground mines. This communication scheme uses Amplitude Shift Keying type of modulation. One of the main advantages of this scheme is that it does not need any line-of-sight communication.

The Smart environment always provides real world data by sensing them or through any interfaces. In this design, the sensing system is realized by various sensors acting like a network. These sensors are used to sense and collect details of various parameters of the working environment.

II. LITERATURE SURVEY

In 2000, Matine Lienard and Pierre Degauque proposed ―Natural Wave Propagation In Mine Environments‖ [2]. In this, natural wave propagation is used for mobile localization and channel nature determination. They have considered the frequency range of 150 MHz to 900 MHz. The ray launching technique is used for arbitrary shape galleries. In 2001, Y.P.Zhang, G.X.Zheng and J.H.Sheng proposed ―Radio Propagation at 900 MHz in Underground Coal Mines‖ [3]. The physical nature of mine environments is categorized for characterizing the radio channels. They have considered the frequencies 200 MHz, 415 MHz and 1000 MHz. For describing channel characteristics, hybrid tunnel model is used. In 2004, Jose Vasquez, Victor Rodriguez and David Reagor proposed ―Underground Wireless Communications Using High-Temperature Superconducting Receivers‖ [4]. By using high temperature superconducting (HTS) materials and digital signal processors (DSP), they have developed communication system of low frequency range. A compressed audio signal of 500 Hz bandwidth is sent on a carrier of 4000 Hz frequency, by QPSK modulation through earth. In 2005,

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Graham.M.Brooker, Steve Scheding, Mark.V.Bishop and Ross.C.Hennessy proposed ―Development And Application Of Millimeter Wave Radar Sensors For Underground Mining‖ [5]. This is suitable for the measurement application compared to other techniques like microwave radar, acoustic and laser technologies. Among many of the millimetre techniques, Frequency Modulated Continuous Wave (FMCW) method is selected due to its simplicity. In 2008, Angus F. C. Errington, Brian L. F. Daku, and Arnfinn Prugger proposed ―Initial In–Mine Position Estimation Using RFID Tags‖ [6]. They proposed Simultaneous Localization and Mapping (SLAM) and used Least Squares approach for position estimation. In 2009, Karl Nienhaus and Martin Hahn proposed ―Wireless Sensing Applications in the Mining and Minerals Industry‖ [7]. They proposed redundant communication system like wireless local area network. It could achieve high quality data through the redundant communication system. In 2010, Zhigao Liu, Chunwen Li, Qingqing Ding and Danchen Wu proposed ―A coal mine personnel global positioning system based on wireless sensor networks‖ [8]. They proposed global positioning system and wireless sensor network for the communication and coverage inside the long tunnel like underground mines. In 2012, Huping Xu, Feng Li, and Yancheng Ma, proposed ―A ZigBee-based miner Localization System‖ [9]. They proposed miner localization and monitoring using Zigbee technology and wireless sensor network.

III. EXISTING METHODS AND DRAWBACKS

There are many existing techniques for the communication inside the underground mines. The communication schemes include GSM, GPS, RFID, Zigbee, Radar sensor network, etc. These techniques are used for the communication inside and outside the underground mines. By integration with other techniques, the efficiency of the selected method can be improved. Each technique has its own advantages and disadvantages.

The Global Positioning System is the most probably used communication system for large coverage area. It is also used for localization and positioning. The advantage of GPS is its long coverage area and the disadvantage is that it is less efficient for underground mines.

The GSM communication technique also covers a large area. Even though it is used in underground mines, communication delay stands as a major disadvantage. Hence this technique is found to be less effective for a critical environment like underground mines.

The use of RFID technology is far better for positioning or localization of underground mine workers and the locomotive devices inside mines. Unlike IR, it does not need Line-of-Sight communication. Even though RFID is technically sound in tracking systems, it is found that the maintenance of RFID tags is a drawback.

Radar sensor technology is more effective for underground mine measurement, which includes the measuring of environment parameters. The sensors are not much affected by the dust. It can penetrate into any type of harsh environment of dust and mist. But the main difficulty which makes it lagging is the high cost needed for the equipments and maintenance.

The Zigbee technology is widely used technique for underground communication. Even though it is a low power and user friendly technique, its cost is more. Due to the widely usage, its cost is trying to be reduced nowadays.

IV. PROPOSED METHOD AND WORKING

The possibility of drastic change in the underground environment can lead to hazards to the miners working inside the mines. Hence there is minimum measure of safety provided. In order to improve the safety of the mine personnel a better sensing system as well as localization system should be developed. These can be realized by an appropriate communication scheme. Radio frequency technology can be proposed to be a better communication scheme inside underground mines. In order to sense and measure the parameters inside the dynamically varying environment parameters Wireless sensors can also be introduced. Now the integration of Radio frequency technology and wireless sensor network can improve the safety inside the mines.

A real time localization system can be achieved via Radio frequency technology and a real time sensing system can be resulted via wireless sensor network at the meanwhile. These two technologies can be used to design a helmet, which can be referred as ―Smart Helmet‖. The Smart helmet consists of various sensors, microcontroller, RF module, etc.

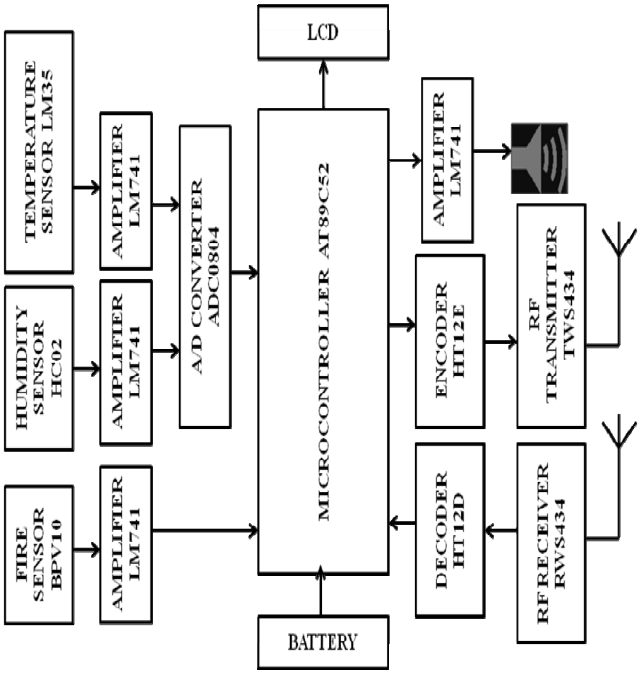


Fig. 1 Smart helmet module

The RF module can be RF transmitter or receiver (or both). It is used to communicate with outside world in order to improve the safety. While designing the prototype, a control room module can be designed to represent the outside world. The various sensors senses the environmental parameters like temperature, humidity, gaseous presence, etc and the measured sensor data is transmitted to the control room to take necessary precautions and warnings for the safety inside the mines

The localization of the mine workers is realized by designing localizer module. The localizer module consists of an encoder, and an RF transmitter.

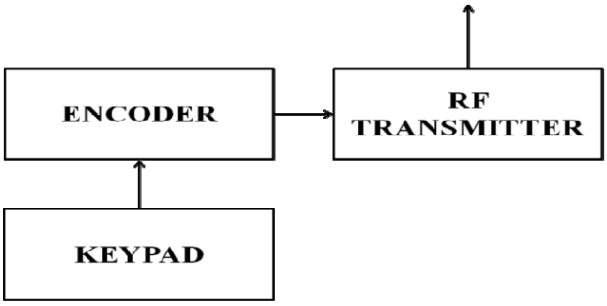


Fig. 2 Localizer module

The encoder of the localizer module is to encode the address of the location of worker inside the underground mine. RF transmitter continuously transmits RF waves in order to detect the presence of the worker.

The design flow of the working of the proposed system is given in Fig.3.

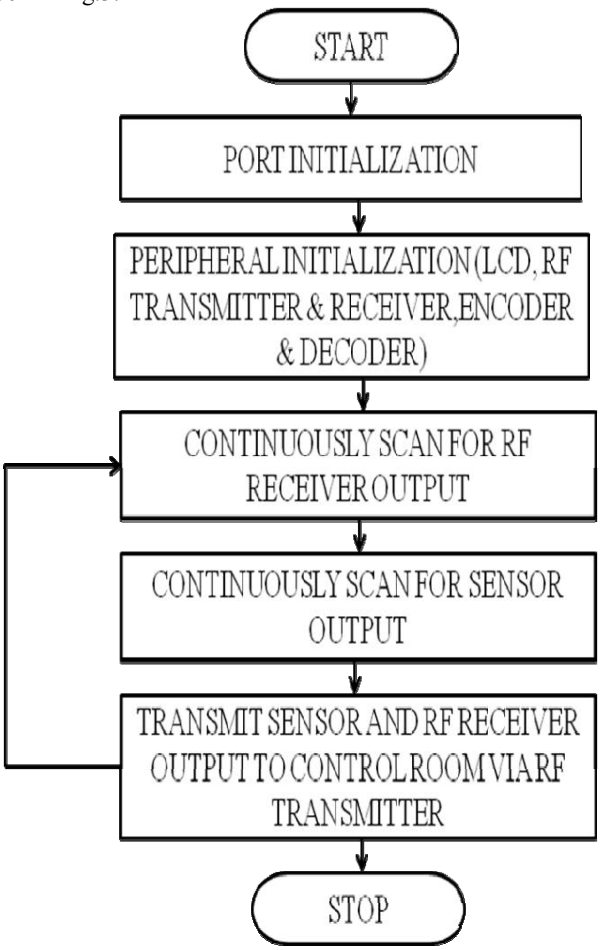


Fig. 3 Design flow

The diagram illustrates that the ports of the microcontroller are initialized. The external peripherals include decoder, encoder, RF transmitter, RF receiver, and LCD. These peripherals are then initialized. The system continuously scans for RF signals and sensor output data. The RF receiver output and the sensor output data are transmitted to the control room for taking enough measures, just like warning systems.

The hardware module includes Smart helmet module, localizer module and control room module. The smart helmet module consists of various sensors to detect and measure various parameters like humidity, temperature, fire, etc. From the knowledge of these parameters, necessary precautions can be taken to prevent hazards occurring inside the mines.

The localizer module contains an RF transmitter for sending RF waves in order to detect the presence of the miners in the mine environment. When the localizer module detects any miner (via helmet module), it sends the data regarding the location of the miners. As a result, the miners trapped during a hazard inside the mines can be rescued. If at any case of damage of the helmet, we can trace the particular miner using the last location updated.

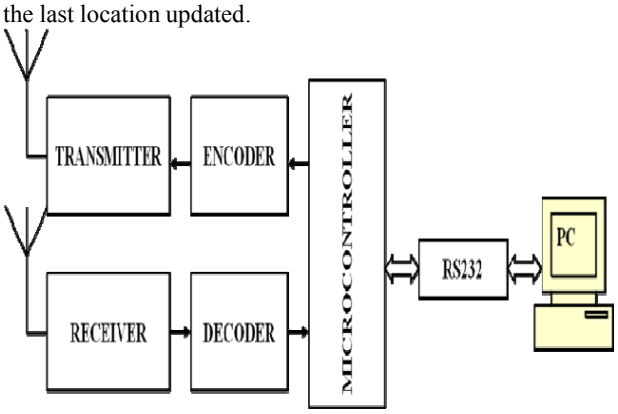


Fig. 4 Control room module

The control room module consists of RF transmitter and receiver, encoder and decoder, microcontroller, RS232 serial interface, and PC. It is given in Fig.4. The RS232 is used for the serial communication between the microcontroller and the PC.

The software implementation can be done by using Proteus VSM software, Keil Microvision software, and Visual basic software. The Proteus VSM software is used to design the circuit layout for the hardware module. Proteus VSM uses ISIS schematic capture software to provide the environment for design entry and development. The ISIS software combines ease of use with powerful editing tools. It is capable of supporting schematic capture for both simulation and PCB design. Designs entered in to Proteus VSM for testing can be net-listed for PCB layout either with Proteus PCB Design products or with third party PCB layout tools. ISIS also provides a very high degree of control over the drawing appearance, in terms of line widths, fill styles, fonts, etc. These capabilities are used to provide the graphics necessary for circuit animation. The Keil Microvision software is used to

program the microcontroller. Like other software, Keil's µVision IDE provides a powerful, easy to use and easy to learn environment for developing embedded applications. They include the components you need to create, debug, and assemble your C/C++ source files, and incorporate simulation for microcontrollers and related peripherals. The RTX RTOS Kernel helps to implement complex and time-critical software. The Visual basic is used for the serial communication between microcontroller and the PC. Visual Basic is a third-generation event-driven programming language and integrated development environment (IDE) from Microsoft. A programmer can create an application using the components provided by the Visual Basic program itself.

V. RESULT AND DISCUSSIONS

The ―Smart Helmet‖ is a real-time sensing and localizing system, which is used to ensure the safety of the mine-workers. The real time sensing is enhanced by various sensors for measuring temperature, humidity, and pressure. The real-time localizing system is enhanced by the radio frequency module by continuously transmitting RF waves. The PCB layout of the main board of the helmet module is shown in Fig.5.

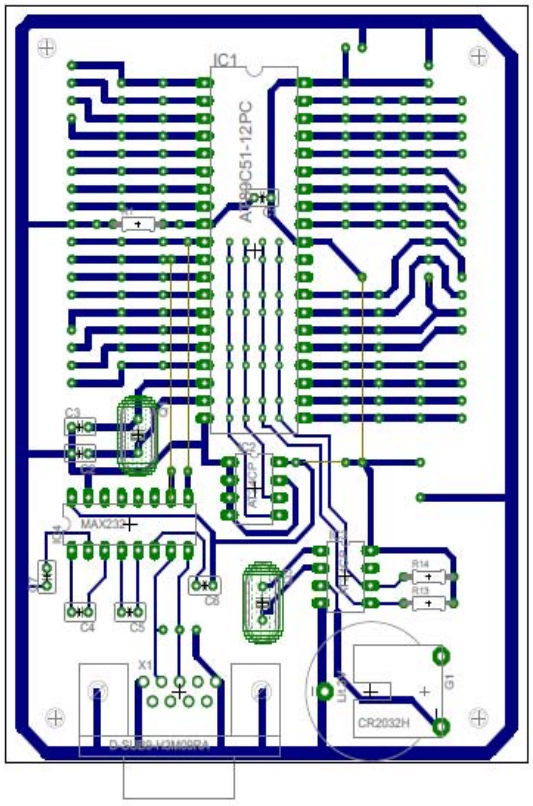


Fig. 5 PCB layout of helmet module

The design and implementation of the prototype of Smart Helmet results in the hardware module as shown in Fig.6.



Fig. 6 Smart Helmet

The sensors attached to the helmet module measure the values of parameters like temperature, humidity and fire detection. The measured values are sent to the control room, where the necessary actions are taken for ensuring the security of mine-workers. The localizer module detects the presence of mine workers and sends their location information to the control room via RF transmitter.

The intelligent warning system through LCD display and alarm provides mine-workers to take safety measures. The alarm system provides warning if the values of parameters like temperature and humidity go beyond their threshold values. The Smart Helmet system can thereby prevents the disasters in the underground mines across the world.

VI. CONCLUSION

Safety has long been a concern in the mining business especially in sub-surface mining. While mining today is substantially safer than it was in the previous decades, mining accidents still occur. Government figures indicate that 5,000 Chinese miners die in accidents each year. Mining accidents continue worldwide, including accidents causing dozens of fatalities at a time, such as the 2007 Ulyanovskaya Mine disaster in Russia, the 2010 Upper Big Branch Mine disaster in the United States, and the 2009 Heilongjiang mine explosion in China. There are many safety gears are available in the market. But SMART HELMET stands versatile among them with its auspicious design and possibilities. Because compared to other safety equipments it helps in continuous monitoring of a mine worker.

By the help of smart helmet variation in surrounding conditions can be monitored and necessary safety precautions can be taken. It also provide a technique for tracking the position of the worker which enables the rescue team to provide immediate help in adverse conditions. As a future enhancement, the control room module can be designed and implemented for the mentioned purposes. The safety provided by the smart helmet can be improved in future by the Integration of more sensors inside the helmet.

The performance of smart helmet can also be increased by using more improved power source which is capable of providing power supply for a very long time. Such a power source will decrease the chance of danger as it ensures continuous protection for an extended period of time. The Smart Helmet system can thereby prevents the disasters in the underground mines across the world. The Rf technology is a low cost technology compared to other communication schemes. Thus Smart Helmet using Radio frequency technology is the Saviour of Underground miners across the world.

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